

DS 593 Final Report: PII Redaction Fine-Tuning

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Introduction

We all care about our health data, but handling who is able to see study data is a delicate matter. Protecting participant health data is a fundamental responsibility in clinical research. This project builds a PII (Personally Identifiable Information)-redaction fine-tuned LLM aimed at redacting PII from participant research visit transcripts.

Many labs, including the one I work for, prioritize data sharing, yet this poses a significant risk in accidentally exposing identifying information. One type of data that is invaluable to cognitive research is voice recordings due to the ease of collection and the interest in analyzing acoustic and linguistic features. However, neither the recordings nor the transcripts themselves may be released to the research community without very careful review due to PII risks. An LLM that could reliably redact any PII from transcriptions could massively accelerate data sharing among research communities.

The core challenge of this task is that errors carry substantial consequences. A missed piece of PII, represented here as a false negative, risks exposing a participant's identity. For this reason, the model must achieve near-perfect recall to be viable for real-world use in sensitive research contexts. If recall falls short, the model cannot be safely deployed without human review, negating much of its value. An additional challenge specific to neuropsychological exams is the story recall test, in which participants recount details of a standardized story that include a character name and a location. When redacting documents, an LLM could plausibly over-redact these story elements, mistaking fictional characters for real participants. This potential for false positives in a structured clinical context was a hypothesis I sought to investigate.

Approach

Model Selection

I chose to fine-tune Phi-4, a 14-billion parameter model developed by Microsoft. While smaller than many LLMs, Phi-4 has demonstrated strong performance on reasoning-intensive tasks, which suggested it would be well-suited to the PII redaction task. Its smaller size also makes this work more easily replicable, even with compute constraints. I chose to fine-tune using Low-Rank Adaptation (LoRA), a parameter-efficient technique that updates a small number of additional weights rather than modifying the full model, significantly reducing memory requirements during training. I paired this with the use of Unsloth, an open-source library that provides optimized LoRA training kernels for large language models. Unsloth reduces VRAM usage and increases training speed significantly compared to standard HuggingFace training, without reducing in model quality. Again, this design choice was particularly aimed at making this work accessible to those using Colab or other affordable GPU providers.

Fine-tuning is a good fit for this PII redaction task for several reasons. PII redaction is a highly specific, well-defined task with a stable output format. What constitutes PII under specific lab guidelines is clearly defined and changes infrequently, making it well-suited to a fine-tuned model rather than a general-purpose one. Fine-tuning also reduces token usage at inference time, which is not a cost concern for a locally deployed open-weight model but contributes to efficiency.

Dataset

I decided to work exclusively with synthetic data throughout this project so that the entire pipeline could be shared openly without risk of exposing real participant information. This decision makes the work fully replicable. However, I would like to acknowledge that synthetic data cannot fully capture the quality or diversity that real data would provide.

My primary training dataset was the ai4privacy HuggingFace PII Masking 300k dataset, which contains 300,000 samples spanning 27 PII categories across 749 distinct subjects. I preprocessed this dataset to retain only English-language examples, since the neuropsychological exams conducted in my lab are English-only. I mapped the dataset's original PII tags to a reduced set based on HIPAA's definition of Protected Health Information (PHI), discarding tag categories that fell outside the HIPAA framework. From the processed dataset, approximately 3,000 examples were used for fine-tuning.

To evaluate model performance, I used a two-tiered approach. First, I used 1500 examples pulled from the ai4privacy validation split of their dataset, which closely resembled the structure and content of the training data. To target my desired use case, I also created a synthetic dataset of 100 neuropsychological exam transcripts using Claude Sonnet. I kept this dataset small enough to manually review every example, ensuring quality and label accuracy, while large enough to produce meaningful evaluation metrics. This dataset served as my gold-standard evaluation set.

PII Tag Schema

PII tags were derived from HIPAA's 18 Protected Health Information identifiers, mapped from the ai4privacy schema. The final tag set used for training and evaluation is shown below.

Tag	Description
NAME	Patient, caregiver, or clinician names
TITLE	Professional titles (Dr., Mr., Mrs.)
DOB	Date of birth
DATE	Appointment or visit dates
TIME	Appointment or visit times
EMAIL	Email addresses

PHONE	Phone numbers
ADDRESS	Street addresses
LOCATION	Cities, facilities, institutions
SSN	Social Security Numbers
UNIQUE_ID	Medical record numbers, patient IDs
LICENSE_NUMBER	NPI or driver's license numbers
IP_ADDRESS	IP addresses
USERNAME	Username or login handles
PASS	Passwords or PINs
SEX	Sex or gender identifiers
CARDISSUER	Credit or debit card issuers

I specifically chose HIPPA's definition of PII because many medical labs pull or extend their definitions from HIPPA's.

Training Setup

Evaluation

I structured evaluation across two tiers, each designed to test a different aspect of model performance.

Tier 1: ai4privacy Held-Out Set

This dataset was created by stratifying a sample of 1,200 examples drawn from the unseen portion of the ai4privacy dataset. This set tests generalization within the training distribution. Because the model was trained on a subset of this dataset, strong performance here is expected but necessary to confirm that the model learned the task rather than overfit. The held-out sample was constructed to ensure a minimum of 75 positive examples per PII tag type, with a 70/30 split between samples containing PII and clean samples.

Tier 2: Synthetic Neuropsychological Transcripts

This dataset comprises 100 manually reviewed synthetic transcripts representing the real target deployment domain. It is aimed to test domain generalization to my target domain, assessing whether the model can redact PII in long-form, unstructured language that was entirely absent from training. This is the primary benchmark for assessing real-world viability.

The baseline for comparison is the non-fine-tuned Phi-4 model evaluated with zero-shot prompting on both sets.

Performance Metrics

Performance on the PII redaction task was evaluated first at the tag level and then at the token level. I initially reported on the tag level because the metrics were easy to generate and it provided an initial point of comparison for the base Phi4 model versus the fine-tuned models. However, it quickly became evident that I need a more fine-grained method for evaluating performance since the presence of a specific PII tag does not indicate that the PII is actually being tagged correctly. For token level evaluation, each individual PII token was counted independently as a true positive, false positive, or false negative. This approach naturally handles cases where the model splits a single entity into multiple tagged spans (e.g., tagging a first and last name separately), which I observed as a common model behavior. Metrics reported include precision, recall, F1 score, and macro F1 across all tag types. Recall is treated as the primary metric given the asymmetric cost of missed PII in a clinical context.

I also applied an additional hallucination check. Any predicted PII token that did not appear in the original source text was flagged as a non-match and excluded from the true positive count. This catches cases where the model generates PII that was never present in the input.

Results and Discussion

Performance on the ai4privacy validation set

Overall, the fine-tuned models performed significantly better than the base Phi4 model across metrics, which were the following:

Model	Precision	Recall	F1
Base Phi4	0.610	0.321	0.421
Fine-tuned 3k	0.943	0.959	0.951
Fine-tuned 20k	0.974	0.981	0.978

Performance on the held-out ai4privacy set was strong, which held with the expectation that a model trained on this distribution would generalize well within it. The model also saw a significant reduction in the hallucination rate with the fine-tuned adapters: the base model hallucinated 282 tokens where the 3k and 20k models hallucinated 10 and 15 tokens respectively.

Performance on the Synthetic NP Transcripts

The fine-tuned models still performed better than the base Phi4 model across metrics; however, all three models struggled to accurately recall all of the relevant tokens. Overall results were the following:

Model	Precision	Recall	F1
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Base Phi4	0.507	0.201	0.288
Fine-tuned 3k	0.863	0.394	0.541
Fine-tuned 20k	0.952	0.132	0.232

Performance on the synthetic neuropsychological transcripts was substantially lower, with errors predominantly being false negatives. This indicated that the model consistently missed PII rather than hallucinating or misclassifying it, which I was able to confirm by checking the logged data. Significantly, the recall for NAME was extremely low, recall being only 0.042 for the 20k model and 0.169 for the 3k model.

Initially, the performance gap was not apparent, as early evaluation was conducted only on the ai4privacy held-out set. When the synthetic NP transcripts were introduced as a second evaluation tier, the performance gap between domains became visible. This underscores the importance of domain-specific evaluation sets for tasks with a well-defined deployment target.

Discussion

The large performance gap between the two evaluation tiers is attributable to a mismatch between the training distribution and the target domain. The ai4privacy dataset consists of short, syntactically simple, declarative sentences in which PII appears in predictable positions. The model learned to recognize these surface patterns effectively. Neuropsychological transcripts, by contrast, embed PII within long, complex clinical speech where names, dates, and identifiers appear in varying positions and formats.

The failure mode is consistent across tag types. For NAME and DATE/DOB, the model appears to miss PII when it is embedded in clinical prose rather than presented as an isolated fact. For SSN, the model likely learned to recognize SSNs in conversational or form-based formats common in the training data, but not in the structured clinical notation (e.g., prefixed by “Pt. ID:” or “SSN:”) that appears in transcripts.

Notably, the story recall test hypothesis was not confirmed as a significant source of errors. To check this, I read through the response output from the fine-tuned and base models, but gathering specific performance metrics around this edge case would be a useful endeavor. False positives on fictional character names and locations were not a dominant failure mode in the NP transcript evaluation. The more pressing issue was under-detection rather than over-detection.

Ethics and Limitations

Ethical considerations are intrinsically tied to this project since it aims to address ethical concerns around participant data privacy. I identified four primary ethical limitations to my current work.

First, fine-tuning a pre-trained model carries a risk of degrading built-in safety features. Testing the fine-tuned model against a set of adversarial prompts to verify that safety guardrails remain

intact was included in the original evaluation plan but was not completed within the project timeline. Any deployment of this model, even in a restricted research context, should include safety evaluation before use. This is noted as a priority for future work.

Second, deploying this model against real participant data requires that inference be conducted on local servers or within a secure compute environment so that PII-containing transcripts never leave institutional infrastructure. For this project, only synthetic data was used throughout, so this was not a concern. However, any extension to real transcripts must prioritize careful data handling as a requirement.

Third, because the model's recall on clinical transcripts is insufficient for unsupervised deployment, using it in a production context without human review would risk PII exposure. The model in its current form should be considered a tool to assist human reviewers, not replace them.

Finally, all data used in this project was synthetic, which makes the work fully replicable and shareable but means that performance estimates may not transfer directly to real clinical transcripts, which are likely noisier, more variable in format, and more linguistically complex than the synthetic examples used here.

Future Work

I initially set out to test several directions that were not completed within the scope of this project. Few-shot and one-shot prompting experiments were originally intended to test whether in-context examples would perform similarly to fine-tuning on synthetic NP transcripts. This comparison would clarify whether fine-tuning is strictly necessary for domain adaptation, or whether a well-prompted base model achieves comparable performance. This remains an open question.

Safety evaluation was not completed. A structured evaluation using a standardized set of adversarial prompts should be conducted to verify that fine-tuning did not degrade the model's ability to refuse harmful requests. This is a prerequisite for any broader deployment.

Domain adaptation via targeted data augmentation is the most direct path to improving recall on NP transcripts. Adding even several hundred clinical-register examples to the fine-tuning set that include unstructured and long-form speech would likely produce a significant improvement given Phi-4's strong underlying clinical language comprehension from pretraining.

Finally, evaluation on real, de-identified clinical transcripts conducted within a secure compute environment would provide the most meaningful indication of actual viability and should be pursued if the project is extended to a research lab setting.

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